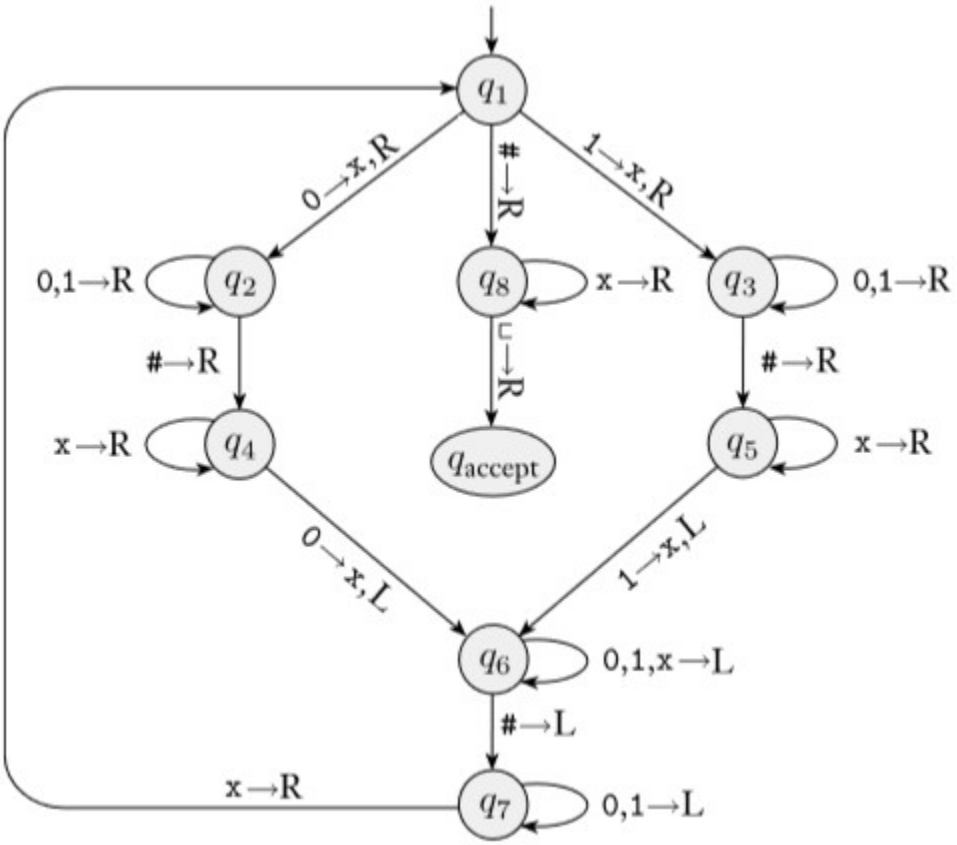
Assignment 3

1/ Given the Turing machine



In each of the parts, give the sequence of configurations that M1 enters when started on the indicated input string.

a. 11.

q_1 11

$\Rightarrow$ x q_3 1

$\Rightarrow$ x 1 q_3 □

No transition is available for q_3 on □, so M1 halts and rejects.

b. 1#1.

q_1 1#1

$\Rightarrow$ x q_3 #1

$\Rightarrow$ x# q_5 1

$\Rightarrow$ x q_6 #x

$\Rightarrow$ q_7 x#x

$\Rightarrow$ x q_1 #x

$\Rightarrow$ x# q_8 x

$\Rightarrow$ x#x q_8 □

$\Rightarrow$ x#x□ q_{accept} □

M1 accepts 1#1

c. 1##1.

q_1 1##1

=> x q_3 ##1

=> x# q_5 #1

No transition is available for q_5 on #, so M1 halts and rejects.

d. 10#11.

q_1 10#11

$\Rightarrow$ x q_3 0#11

$\Rightarrow$ x0 q_3 #11

$\Rightarrow$ x0# q_5 11

$\Rightarrow$ x0 q_6 #x1

$\Rightarrow$ x q_7 0#x1

$\Rightarrow$ q_7 x0#x1

$\Rightarrow$ x q_1 0#x1

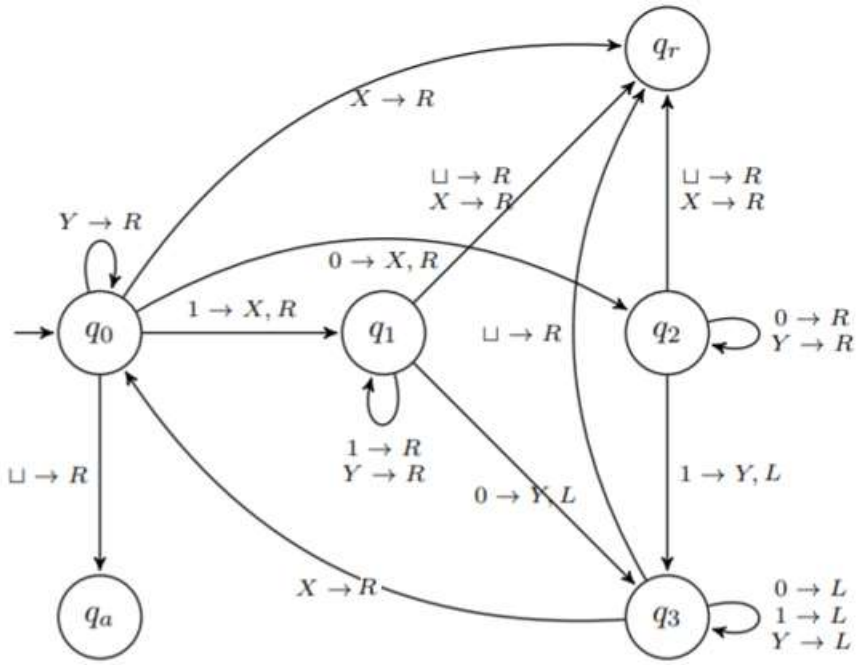
$\Rightarrow$ xx q_2 #x1

$\Rightarrow$ xx# q_4 x1

$\Rightarrow$ xx#x q_4 1

No transition is available for q_4 on 1, M1 halts and rejects.

2/Given the Turing machine M with the following state diagram:



a/Describe the language accepted by M

The Turing machine accepts the string if and only if the number of 0s equals the number of 1s.

It matches every 1 in the string with a corresponding 0 (and vice versa)

The process of matching a zero and its corresponding 1 is: mark zero by X, go right until the head reaches the corresponding 1, mark with Y and go left to mark a new pair (0,1) or (1,0).

When the head reaches the rightmost X, it goes right and reaches state q_0.

b/Give the sequence of configurations for the input string

0011

q_0 0011

$\Rightarrow$ X q_2 011

$\Rightarrow$ X0 q_2 11

$\Rightarrow$ X0Y q_3 1

$\Rightarrow$ X0 q_3 Y1

$\Rightarrow$ X q_3 0Y1

$\Rightarrow$ q_3 X0Y1

$\Rightarrow$ X$q_0$0Y1

$\Rightarrow XXq_2Y1$

$\Rightarrow XXYq_21$

$\Rightarrow XXq_3YY$

$\Rightarrow Xq_3XYY$

$\Rightarrow XXq_0YY$

$\Rightarrow XXYq_0Y$

$\Rightarrow XXYY\square\ q_a\square$

The TM halts and accepts 0011.

0110

q_0 0110

$\Rightarrow X\ q_2\ 110$

$\Rightarrow XY\ q_3\ 10$

$\Rightarrow X\ q_3\ Y10$

$\Rightarrow q_3\ XY10$

$\Rightarrow Xq_0\ Y10$

$\Rightarrow XY\ q_0\ 10$

$\Rightarrow XYX\ q_10$

$\Rightarrow XYX\ q_3Y$

$\Rightarrow XYq_3XY$

$\Rightarrow XYXq_0Y$

$\Rightarrow XYXYq_0\square$

$\Rightarrow XYXY\square\ q_a\square$

The TM halts and accepts 0110.

c/show that string 10011 is not accepted by M:

10011

q_0 10011

$\Rightarrow X\ q_1\ 0011$

$\Rightarrow$ XY $q_3$011

$\Rightarrow$ XY X$q_2$11

$\Rightarrow$ XY XY$q_3$1

$\Rightarrow$ XYXq_3Y1

$\Rightarrow$ XYq_3XY1

$\Rightarrow$ XYXq_0Y1

$\Rightarrow$ XYXY$q_0$1

$\Rightarrow$ XYXYXq_1□

$\Rightarrow$ XYXYX□q_r□

The TM halts and rejects 10011